

Project Title

SECURE MARKSHEET VERIFIER: Tamper-Evident Verification

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Physical marks sheets are still the common credential that students give to interviewers HR recruiters and scholarship committees.. I find there is no quick reliable way for these third parties to confirm authenticity. A printed mark can be easily altered. For example changing a 78 to a 99. And verifying it now means I have to contact the university or request database access, both slow and impractical. This also brings privacy concerns because verifying one students record should never expose another students data. At the time organizations need instant trust in the documents they receive without waiting for institutional back-and-forth. Binary Hash Trees offer an elegant solution: they let a university anchor every students record into a single tamper-proof root hash enabling any third party to verify one record in $O(\log N)$ time using only a small audit path with zero exposure of other students' data. Combining this with photo OCR bridges the gap, between paper and cryptographic verification. Motivating this project.

State of the Art / Current solution

All existing credential verification systems have their limitations. Manual Verification involves the third party calling or writing to the registrar of the university and waiting for days before getting a response, thus being inefficient and difficult to scale. Centralized Digital Portals like DigiLocker and National Academic Depository store verified digital mark sheets but work only when the student has a digitized version and the university is registered on that platform; a printed mark sheet cannot be verified in this case. QR codes embedded within marksheets refer to an online record which may still get bypassed if the QR code is duplicated from a legitimate marksheet. Block Chain-based Degree Verification, adopted by a few universities across the globe, is immutable, but needs extensive network and consensus system which makes it expensive for a single organization. Database Access involves revealing the entire database of students to the third party, thus posing privacy concerns.

Project Goals and Milestones

Goals:

1. Construct a binary Merkle Hash Tree using C++, where all records of university students will be anchored to a single Root Hash.
2. Construct a hash engine called CustomHash that combines FNV-1a and DJB2 from scratch, without using any external cryptographic library, based on DSA principles.
3. Integrate a process to recognize text from a photograph of physical marksheet for extraction of roll number, subject codes, and marks.
4. Construct a verification engine that will do two jobs, i.e., checking of the Merkle proof in $O(\log N)$ and also field-wise verification to locate the tampered subjects' marks.
5. Privacy preserving verification such that a third party can verify a single student's record without having access to the records of any other student.

Milestones:

- M1: CustomHash module implementation with deterministic hash generation with FNV-1a + DJB2 algorithm.
- M2: Bottom up Merkle Tree creation along with Root Hash generation using the whole student data set.
- M3: Logarithmic time complexity Merkle Proof creation along with independent proof verification mechanism.
- M4: Integration of photo OCR engine along with the marksheet text parser.
- M5: End to end verification engine which provides final verdicts in CSV logs tested against authentic, tampered, and counterfeit sample marksheets..

Project Approach

The architecture uses a two-pronged method. University end: student records (roll number, subjects, marks) are fetched from csv file, parsed, and hashed using a custom made FNV-1a + DJB2 hashing engine CustomHash), which is hand-written in C++. Leaf hashes are used to build a binary Merkle tree, resulting in creation of a single Merkle Root Hash per batch which is published only once and stored in merkleledger.csvfile.

Verification end: the interviewer or the scholarship evaluator photographs the applicant's mark sheet. The OCR engine (developed using Windows Media OCR) reads out the roll number and the marks in every subject, which are parsed again and hashed. The verification engine runs two verifications: a logarithmic time Merkle proof check of the hash against the university's Merkle root, and a field by field audit of every mark against the official data. Depending upon the results, a verdict is issued.

System Architecture (High Level Diagram)

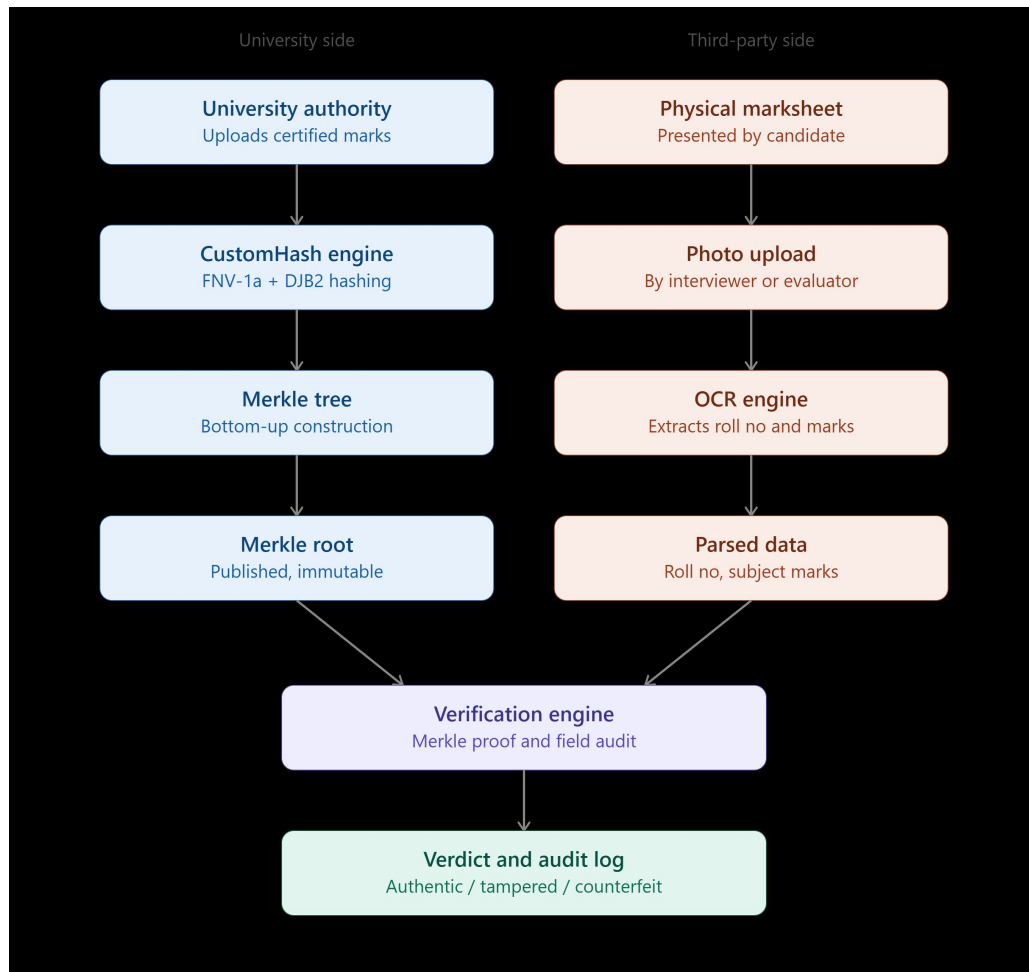


Figure 1: System Architecture

Project Outcome / Deliverables

The project results in the completion of a complete, tamper-resistant marksheet verification framework using only C++. The project uses a self-developed hash functions to create a bottom-up Merkle Hash Tree in which all records of university students will be embedded into one root hash. A $\log(N)$ Merkle proof generator for the project allows independent verification of any student's record without revealing the database at all. The project uses a photo-based OCR engine, developed based on Windows Media OCR to extract the roll number and marks of subjects from the photographed physical marksheet and then verifying using a verification engine for both cryptographic Merkle proof verification and field-by-field mark audit. All official records, published hashes, and verification sessions stored in CSV files.

Assumptions

The system makes an assumption that the university has digitized and authenticated student records stored in csv file prior to building the Merkle Tree. The system makes an assumption that the physical image of the marksheet has been photographed legibly due to the dependency of Optical Character Recognition on legibility. It also makes an assumption that the Merkle Root Hash has been published in a way that is secure and tamper-proof, and that the roll number of the physical marksheet corresponds to the record of the university.

References

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